

SMART INDIA HACKATHON 2026

PROBLEM STATEMENT ID: SIH26003

VERSION 2.0 • ENHANCED MASTER BLUEPRINT

AI-Based Cognitive Gaming & Memory Assistance Platform for Elderly Dementia Patients in NER

An Enhanced Technical Blueprint, Clinical Validation Framework, Offline Architecture & Execution Roadmap for the North Eastern Region of India (MDoNER).

MINISTRY / SPONSOR

MDoNER, Govt. of
India

PLATFORM CODENAME

CogniCare NER
(স্মৃতি-সেতু)

CATEGORY & DOMAIN

MedTech /
Healthcare AI

TARGET DEPLOYMENT

8 NER States +
ASHA Nodes

Official Deep Dive & Critical Analysis of the Initial Guide

The initial guide for SIH26003 established a solid baseline—introducing Contextual Multi-Armed Bandit (Thompson Sampling) difficulty adaptation, the 60–80% "Flow Zone" concept, Composite Cognitive Scoring (CCS), and offline-first data sync. However, when evaluating the solution against real-world geriatric dementia psychiatry in the North Eastern Region (NER), several critical clinical, architectural, and demographic gaps were identified.

Key Insight from ARDSI & LGBRIMH Tezpur Research: In remote hill districts, elderly individuals suffering from dementia cannot be treated as standard mobile gamers. Repetitive cognitive tests cause frustration; dialects vary every 50 kilometers; and family members or ASHA health workers serve as the crucial intermediary interface.

Identified Gaps vs. Enhanced Engineering Solutions

✗ Initial Guide Limitations

- Relies exclusively on abstract puzzles & games, risking patient agitation.
- Assumes elderly patients independently operate complex smartphone apps.
- Ignores "Sundowning" (evening anxiety, restlessness, and disorientation).
- Relies on unverified offline STT (Vosk) for unsupported hill dialects.
- Produces raw 0–100 scores with no mapping to clinical tests (MMSE/MoCA).
- No emergency safe-return protocol for patient wandering disorientation.

✓ Version 2.0 Enhanced Master Blueprint

- **Reminiscence & Nostalgia Therapy:** Photo albums, folk tales & soothing music.
- **ASHA Assisted Mode:** 5-minute guided screening & batch data collection.
- **Sundowning Relief Mode:** Automated amber lighting & ambient flute therapy.
- **Hybrid Audio Architecture:** Pre-cached native dialect voice phrasebooks.
- **Clinical Standard Mapping:** Direct conversion of CCS to MMSE/MoCA stages.
- **Safe Home SOS Card:** One-tap visual address and caregiver geo-alert.

1. Reminiscence & Nostalgia Preservation Therapy (Clinical Priority)

Dementia progressively degrades short-term episodic memory while preserving long-term autobiographical memories until late stages. The enhanced platform introduces an interactive **Reminiscence Storybook & Nostalgia Player**:

- **Family & Memory Album:** Displays tagged photos of family members, village homes, and cultural celebrations with soothing voice prompts ("This is your grandson, Nilav, celebrating Bohag Bihu").
- **Cultural Audio Stimulation:** Curated library of traditional folk songs (Borgeet, Tokari Geet, Cheraw bamboo rhythms, Bihu flute) proven to lower cortisol levels and calm agitated patients.

2. Dedicated ASHA & Community Health Worker "Assisted Screening" Mode

In tribal and rural pockets of Assam, Meghalaya, Nagaland, and Arunachal Pradesh, ASHA workers are the primary healthcare touchpoint. The platform provides a streamlined **ASHA Portal**:

- Administer a standardized 5-minute cognitive check-in across 5 domains.
- Log behavioral observations (sleep quality, appetite, mood changes).
- Store results locally in SQLite / Isar DB with automatic sync upon reaching network coverage.

3. "Sundowning" Evening Calming & Anti-Agitation Protocol

Between 4:00 PM and 7:00 PM (early dusk in the North East), dementia patients frequently experience heightened agitation and confusion. CogniCare automatically activates **Sundowning Calming Mode**:

- Switches to ultra-warm, low-blue-light amber themes (#FFF3E0 / #5D4037).
- Plays gentle natural soundscapes (rain on bamboo roofs, gentle river streams).
- Displays reassuring family voice notes and gentle orientation reminders.

4. Emergency "Where Am I?" / Wandering Safe-Return Card

Disorientation is life-threatening. The top-level interface features a persistent, high-visibility **Safe Home SOS Button** that presents the patient's home village, caregiver contact, and immediate one-tap calling without complex menus.

5. Clinical Correlation: CCS to MMSE / MoCA Diagnostic Stages

To make caregiver reports actionable for neurologists at **LGBRIMH Tezpur** or AIIMS Guwahati, our Composite Cognitive Score (0–100) translates into clinical diagnostic equivalencies:

CCS SCORE RANGE	ESTIMATED MMSE EQUIVALENT	CLINICAL DEMENTIA RATING (CDR)	RECOMMENDED INTERVENTION
80 – 100	27 – 30 (Normal / High Function)	CDR 0 (Healthy Aging)	Preventive cognitive agility games
65 – 79	22 – 26 (Mild Cognitive Impairment)	CDR 0.5 (Questionable / MCI)	Adaptive memory puzzles + routine reminders
45 – 64	15 – 21 (Mild Dementia)	CDR 1.0 (Mild Impairment)	Reminiscence therapy + assisted daily tasks
< 45	< 15 (Moderate to Severe)	CDR 2.0+ (Moderate / Severe)	Soothing audio, caregiver alerts, family photos

6. Realistic Offline Multilingual Voice Phrasebook

Instead of relying solely on heavy cloud APIs or imperfect offline STT models for under-resourced languages, the app bundles **150+ pre-recorded native audio prompts** in Assamese, Meitei (Manipuri), Bodo, Mizo, Khasi, Hindi, Bengali, and English for 100% reliable offline playback.

End-to-End System Architecture & Dual Surface Model

The architecture combines an **Offline-First Mobile/Tablet Interface** for the elderly patient and visiting ASHA workers with a **Real-Time Web Intelligence Dashboard** for family caregivers and medical specialists.

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| COGNICARE NER SYSTEM BLUEPRINT |
|
| 📱 PATIENT & ASHA EXPERIENCE (Mobile/Tab) | 🌐 CAREGIVER & CLINICIAN WEB
PORTAL | | • Large 60dp+ touch targets & tactile UI | • Real-time CCS score
analytics & trends | | • 6 Adaptive Cognitive Games (Flow Zone) | • Multi-
domain spider radar analysis | | • Reminiscence Photo & Folk Audio Album | •
Automatic decline & anomaly alert triage | | • Multi-language voice narration
& prompts | • ASHA community patient roster & notes | | • Gentle chime medicine
& hydration alerts | • 1-Click Clinical PDF/CSV Report export | | • Emergency
"Safe Home" SOS Card | • Remote reminder & medication scheduler |
|
| ⚡ LOCAL ON-DEVICE ML ENGINE (TensorFlow Lite + Contextual Thompson
Sampling) | | • Real-time feature extraction: Accuracy, response latency,
motor jitter, pause count | | • Flow Zone targeting (60-80% accuracy optimal
engagement) | | • Zero latency execution, 100% functional without internet
connectivity |
|
| 🗄️ OFFLINE-FIRST ISAR / SQLITE LOCAL STORE —[Auto-Sync on Network]→
FASTAPI CLOUD | | • Append-only game session telemetry | • PostgreSQL: Users,
RBAC, Profiles | | • Server-wins medication schedules | • MongoDB:
Longitudinal cognitive metrics | | • Pre-cached dialect audio assets | • Cloud
ML: Anomaly detection & analytics |

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Composite Cognitive Score (CCS) Mathematical Formulation

The patient's overall cognitive state is continuously evaluated through a multi-dimensional weighted index:

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# Composite Cognitive Score (0 to 100) CCS = (0.25 * Score_Memory) + (0.20 *
Score_Attention) + (0.20 * Score_Executive) + (0.20 * Score_Visuospatial) +
(0.15 * Score_Language) # Where each Domain Score is calibrated against:
Domain_Score = (0.45 * Accuracy) + (0.25 * Speed_Index) + (0.15 * Consistency)
+ (0.15 * Motor_Precision)

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Culturally Grounded Cognitive Games Catalog (8 NER States)

GAME & DOMAIN	NER CULTURAL THEME	MECHANICS & ELDERLY DESIGN	ADAPTIVE PARAMETERS
 Gamosa & Shawl Match Working Memory	Assam (Gamosa), Nagaland (Naga Shawls), Manipur (Inaphi)	Flip large cards to match traditional handloom patterns and regional motifs.	Grid size (2×2 → 4×4), card flip display duration, decoy motifs.
 Festival Rhythm Recall Episodic Memory	Assam (Bihu Dhol), Mizoram (Cheraw Bamboo), Tripura (Garia)	Listen and watch traditional rhythmic instruments; repeat sequence in order.	Sequence length (3 → 7), tempo speed, auditory-only mode.
 Traditional Daily Routine Executive Function	Tea plucking (Assam), Loom weaving, Morning prayers at Namghar/Monastery	Arrange illustrated daily tasks in natural chronological sequence.	Number of steps (3 → 6), distractor activities, voice hints.
 NER Scenic Discovery Sustained Attention	Loktak Lake (Manipur), Living Root Bridges (Meghalaya), Majuli Island	Identify key cultural landmarks or subtle differences in serene landscapes.	Target size, background clutter, subtle color variations.
 Regional Word & Object Language & Naming	Indigenous flora/fauna (One-horned Rhino, Orchids, Hornbill, Tea leaves)	Identify the image and tap the corresponding word or spoken voice token.	Language choice, phoneme similarity, audio hint availability.
 Bamboo Craft Assembly Visuospatial	Tripura bamboo crafts, Monpa masks (Arunachal), Thangka art (Sikkim)	Drag and snap missing geometric pieces into traditional craft silhouettes.	Piece count (2 → 6), snap tolerance, orientation guidance.

Elderly-Friendly UI/UX Standards & Accessibility (WCAG AAA)

Calming Earth Color Palette

Primary: Bark Brown (#5D4037) & Cream (#FFFDF9)

Success: Forest Green (#2E7D32) & Mint (#E8F5E9)

Alert: Warm Terracotta (#E65100)

Avoids harsh blues, neon highlights, or flashing red alarms that trigger anxiety.

Tactile & Motor-Friendly Interactions

Minimum Target Size: 64 x 64 dp for all buttons.

Typography: 22sp base font, 28sp+ headers, high contrast ratio (7:1+).

Navigation: Maximum 2 levels deep, persistent bottom navigation with icon + text.

Elderly Interaction Principles

- **No Punitive Failure Feedback:** Games never show buzzer sounds or red "Game Over" banners. Mistakes trigger gentle voice encouragement ("Let's try this together!").
- **Voice-First Parity:** Every screen provides an audio speaker icon to read prompts aloud in the patient's selected mother tongue.
- **Zero Time-Pressure Mode:** Timers are purely informational for analytics and never kick the patient out of a game unexpectedly.

SIH Grand Finale Presentation & Demonstration Strategy

Judges from **MDoNER** and medical advisory boards look for genuine empathy, real technical depth, and tangible feasibility in remote North Eastern conditions.

TIME (7 MIN PITCH)	SEGMENT & VISUAL FOCUS	KEY NARRATIVE & DIFFERENTIATOR
0:00 – 1:00	Emotional Hook: Story of Lakhimi (72, Tezpur)	Highlight NER statistics: 5M+ Indian dementia patients, remote terrain, linguistic diversity, zero localized solutions.
1:00 – 3:00	Live Patient App Demo (Airplane Mode)	Show live Assamese voice narration, play Gamosa memory match, show Thompson Sampling adapting live without internet.
3:00 – 4:30	Caregiver & ASHA Portal Walkthrough	Show multi-patient dashboard, cognitive spider chart, MMSE correlation, and automated decline alert triage.
4:30 – 5:45	Architecture, Security & Sync Engine	Demonstrate reconnecting to Wi-Fi: instant conflict-free sync from local SQLite to cloud PostgreSQL.
5:45 – 7:00	Roadmap, LGBRIMH Pilot & Scalability	Phase 1: Pilot with LGBRIMH Tezpur; Phase 2: ASHA roll-out via NHM; Phase 3: Pan-India regional packs.

CogniCare NER (স্মৃতি-সেতু) — Empowering Dignified Cognitive Care for the North East • SIH 2026